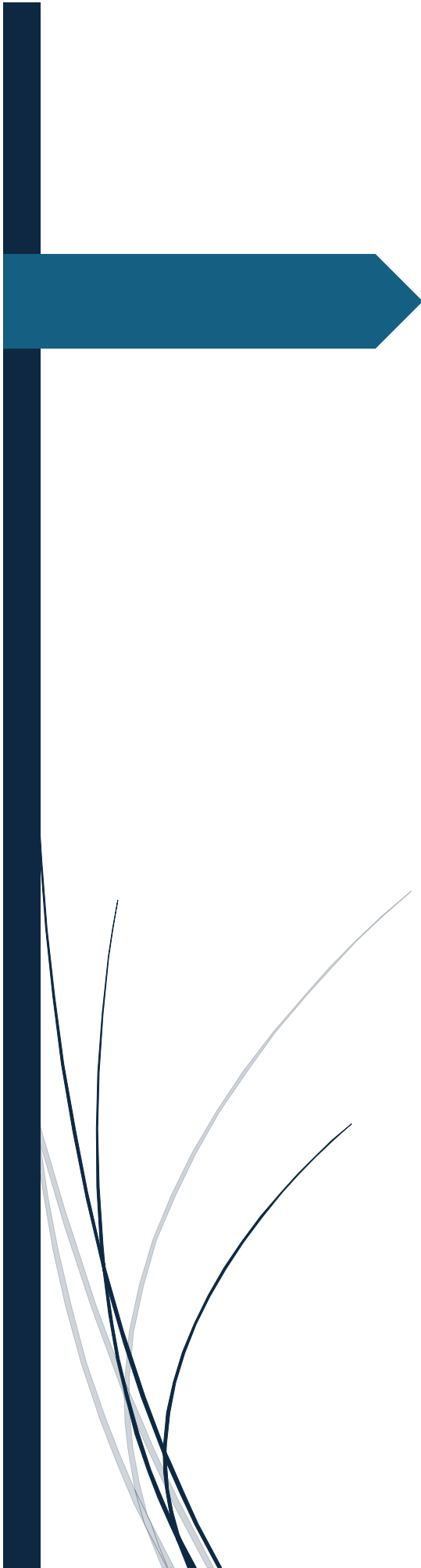




Assignment 1

Tools for Troubleshooting Theory

1. List three real-life scenarios where a multimeter would be used to troubleshoot a computer or smartphone issue, and explain in one line how the multimeter helps in each case.

- **Three Real-Life Uses of a Multimeter:**

1. Computer not turning on:

A multimeter can check the voltage from the power supply and help find out if the PSU is giving enough power.

2. Phone not charging:

A multimeter can check the charging cable, charging port, and battery voltage to find where the charging problem is.

3. Computer overheating:

A multimeter can check the voltage going to the cooling fan to see if the fan is receiving power and working correctly.

2. Watch a short YouTube video demo of a POST card in action (search for 'POST card motherboard test'), then write a brief summary (3-4 lines) describing what the POST card displays and what problem it helped identify in the video.

➤ The video shows how a **POST card** is used to test a computer motherboard. the card is connected to the motherboard and shows **numbers and codes** when the computer starts.

These codes tell us if the computer is starting correctly or if there is a problem. the POST card can help find problems with the **RAM, CPU, BIOS, or motherboard**.

In the video, the POST code helps the user understand where the computer startup problem is.

3. Download and install any free software diagnostic tool (such as HWMonitor, Speccy, or CrystalDiskInfo), run it on your laptop or desktop, and take a screenshot of the main dashboard showing system health details. Constraint: Blur or crop out any personal information before submitting.

Speccy

File View Help

Summary

- Operating System
- CPU
- RAM
- Motherboard
- Graphics
- Storage
- Optical Drives
- Audio
- Peripherals
- Network

Operating System
Windows 10 Home Single Language 64-bit

CPU
AMD Ryzen 7 7435HS 62 °C
Rembrandt 6nm Technology

RAM
16.0GB Dual-Channel DDR5 @ 4791MHz (40-39-39-77)

Motherboard
ASUSTeK COMPUTER INC. FA506NCG (FP7) 60 °C

Graphics
Generic PnP Monitor (1920x1080@144Hz)
NVIDIA GeForce RTX 3050 Laptop GPU (ASUSTek Computer Inc) 45 °C
AMD Radeon(TM) 680M (ASUSTek Computer Inc) 46 °C

Storage
476GB NVMe Micron_2400_MTFDKBA512QFM (Unknown (SSD)) 40 °C
931GB Western Digital WDC WD10SPZX-80Z10T0 (SATA (SSD)) 38 °C

Drive	Capacity	Real Size	Health	Temperature	Type
Local Disk (C:)	476 GB	512,110,190,592 bytes	Good (100%)	40 °C	SSD
Local Disk (D:)	931 GB	1,000,202,273,792 bytes	Good (100%)	38 °C	SSD

Optical Drives
No optical disk drives detected

Audio
Realtek High Definition Audio

4. Compare the use of a multimeter, POST card, and software diagnostics by creating a table with three columns (Tool, what it checks, Example Problem Detected) and fill in at least one example for each tool.

Tool	What it checks	Example Problem
Multimeter	Checks electricity and voltage.	Bad battery or power supply
POST Card	Checks the motherboard when the PC starts.	RAM or motherboard problem
Software Diagnostic	Checks the health of computer parts.	High temperature or disk problem